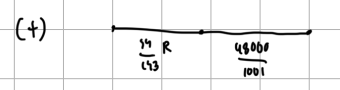
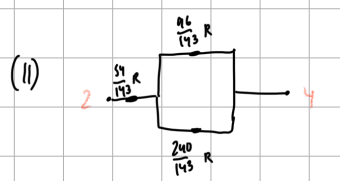
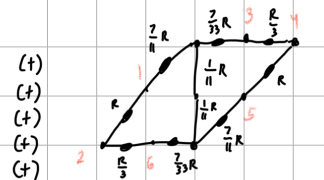
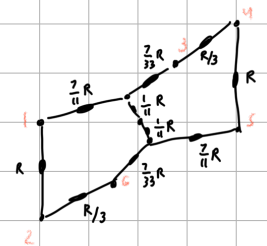
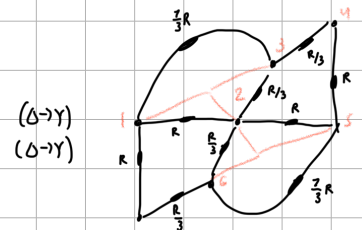
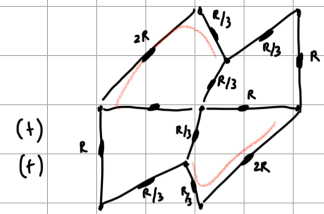
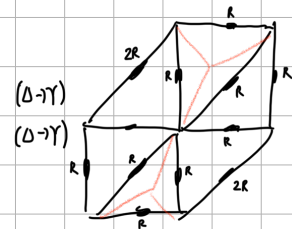
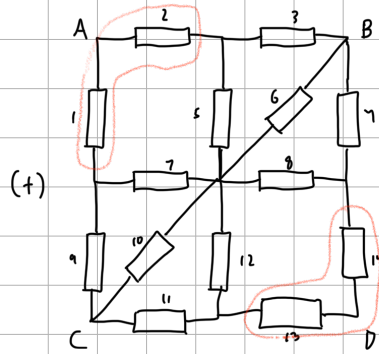


①



$= 85.714 \Omega$

For convenience, we will simplify the circuit using the following "operations":

($\Delta \rightarrow Y$) Delta-Wye

(+) Series

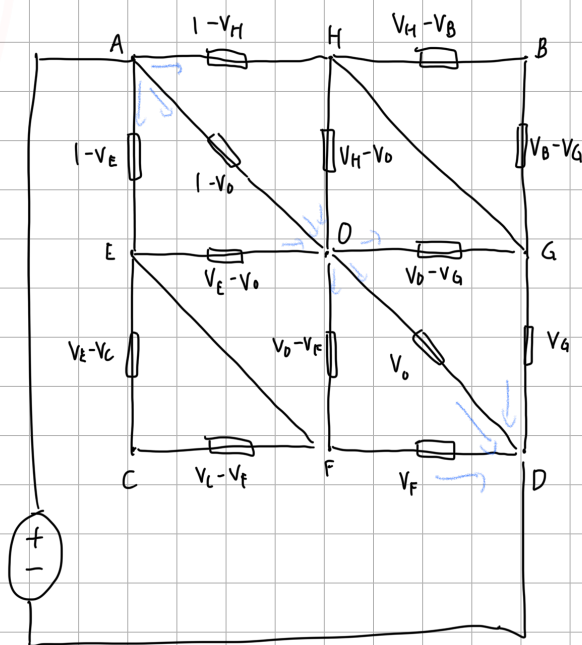
(II) Parallel

(involved resistors are highlighted with red color)

no operation associated implies a redraw for clarity

Multiple operations can happen in one iteration, as long as all values used are from the past iteration and there could be no ambiguity (e.g. the result is the same no matter the order you apply the operations).

2026/09/08



Nodal analysis

$$I_{xy} = \frac{V_x - V_y}{R}$$

Since every current has denominator, let's drop it for the sake of brevity

$$\sum I_{xy} = 0$$

$$\sum \frac{V_x - V_y}{R} = 0$$

$$\sum V_x - V_y = 0$$

Set $V_A := 1 \text{ V}$
 $V_D := 0 \text{ V}$

Note that the circuit is symmetric along $y=x$ and $y=-x$.

That is,

$$V_E = V_H \quad V_C = V_F$$

$$V_G = V_F \quad V_B = V_H$$

$$\Rightarrow V_E = V_G = V_F = V_H \quad V_B = V_C$$

reduce all to V_E

voltage wire

①	$V_F + V_G + V_D$	$= 3 - V_H - V_E - V_D$
D	② $(1 - V_D) + (V_H - V_D) + (V_E - V_D)$	$= (V_D - V_G) + (V_D) + (V_D - V_F)$
G	③ $(V_B - V_G) + (V_D - V_G)$	$= V_G$
F	④ $(V_C - V_F) + (V_D - V_F)$	$= V_F$

reduce all to V_E

①	$2V_E + V_D$	$= 3 - 2V_E - V_D$
②	$1 + 2V_E - 3V_D$	$= 3V_D - 2V_E$
③	$V_B + V_D - 2V_E$	$= V_E$
④	$V_C + V_D - 2V_F$	$= V_F$

Hence, current is

$$I_A = \frac{(V_A - V_H) + (V_H - V_E) + (V_H - V_D)}{R}$$

$$= \frac{\frac{1}{2} + \frac{1}{2} + \frac{1}{2}}{R}$$

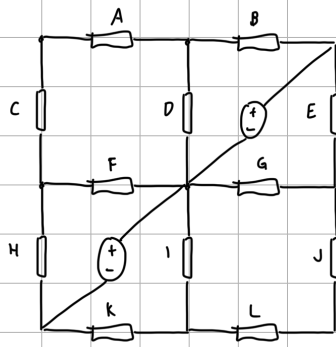
$$I_A = \frac{3}{2R}$$

$$R_{AD} = \frac{1-0}{I_A} = \frac{1}{1.5/R} = \frac{2}{3}R = \boxed{133.333 \Omega}$$

①	$4V_E + 2V_D$	$= 3$
②	$-4V_E + 6V_D$	$= 1$

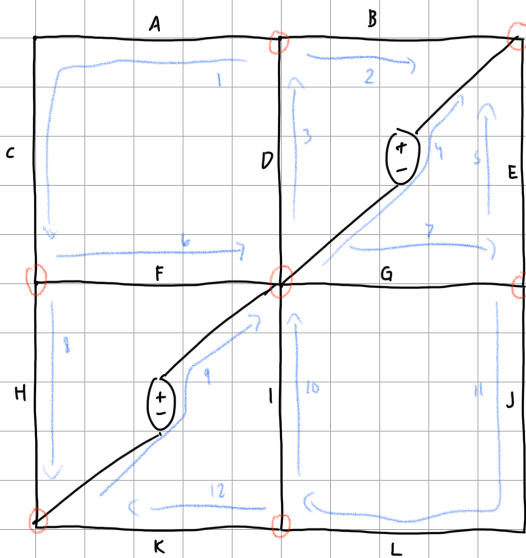
$$\Rightarrow \begin{matrix} 4V_D \\ V_D \end{matrix} = \begin{matrix} 4 \\ 1/2 \end{matrix}$$

so $\begin{matrix} 4V_E + 1 \\ 4V_E \\ V_E \end{matrix} = \begin{matrix} 3 \\ 2 \\ 1/2 \end{matrix} = V_G = V_F = V_H$



1 2 3 4 5 6 7 8 9 10 11 12 V

KJR



$$I_1 = I_6 + I_8$$

$$I_4 + I_2 + I_3 = 0$$

$$I_8 + I_{12} = I_9$$

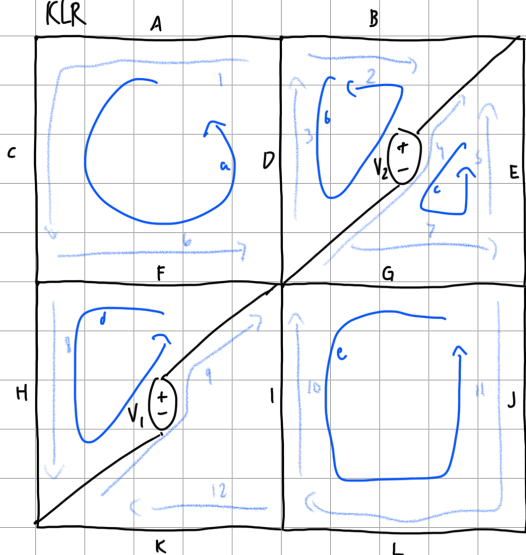
$$I_{11} = I_{10} + I_{12}$$

$$I_7 = I_5 + I_{11}$$

$$I_3 = I_1 + I_2$$

$$I_9 + I_6 + I_{10} = I_3 + I_4 + I_7$$

KLR



$$(a) -I_2 R_A - I_6 R_F - I_3 R_D = 0$$

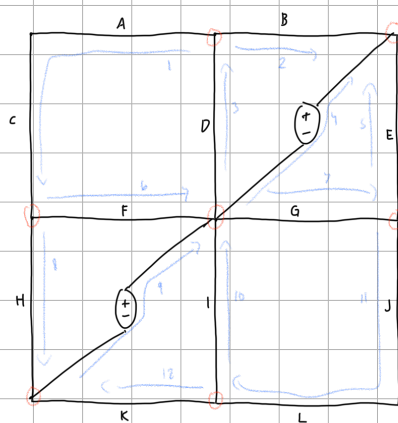
$$(b) +V_2 + I_2 R_B + I_3 R_D = 0$$

$$(c) -I_7 R_G - I_5 R_E - V_2 = 0$$

$$(d) -I_8 R_H + V_1 + I_6 R_F = 0$$

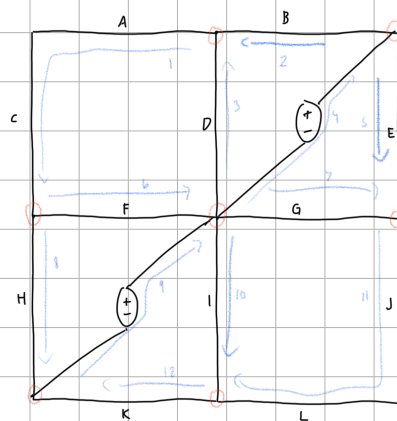
$$(e) +I_{10} R_I + I_{11} R_L + I_{11} R_J + I_7 R_G = 0$$

KJR

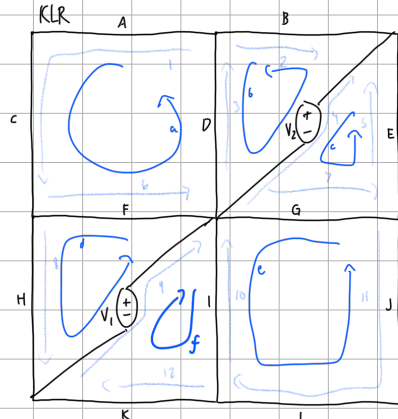


$$\begin{aligned} l_1 &= l_6 + l_8 \\ l_4 + l_2 + l_5 &= 0 \\ l_8 + l_{12} &= l_9 \\ l_{11} &= l_{10} + l_{12} \\ l_7 &= l_5 + l_{11} \\ l_3 &= l_1 + l_2 \\ l_9 + l_6 + l_{10} &= l_3 + l_4 + l_7 \end{aligned}$$

Corrected directions :



KLR



$$\begin{array}{lcl}
 (a) & -I_1 R_A + R_F - I_6 R_F - I_3 R_D & = 0 \\
 (b) & +V_2 + I_2 R_B + I_3 R_D & = 0 \\
 (c) & -I_7 R_G - I_5 R_E - V_2 & = 0 \\
 (d) & -I_8 R_H + V_1 + I_6 R_F & = 0 \\
 (e) & +I_{10} R_I + I_{11} R_L + I_{11} R_J + I_7 R_G & = 0 \\
 (f) & -I_{12} R_K + V_1 + I_{10} R_I & = 0
 \end{array}$$

R	A,C	B	D	V	E	F	G	H	V	I	J,L	K	
I	1	2	3	4	5	6	7	8	9	10	11	12	V

[illegible]

l_1	$= l_6 + l_8$
$l_4 + l_2 + l_5$	$= 0$
$l_8 + l_{12}$	$= l_9$
l_{11}	$= l_{10} + l_{12}$
l_7	$= l_5 + l_{11}$
l_3	$= l_1 + l_2$
$l_9 + l_6 + l_{10}$	$= l_3 + l_4 + l_7$

- | | | | | |
|----|-------------------------------|--|--|--|
| 2 | - R with highest current | | | |
| 4 | - I | | | |
| 5 | - R with lowest current | | | |
| 6 | - I | | | |
| 7 | - R with highest voltage drop | | | |
| 9 | - V | | | |
| 9 | - R with lowest voltage drop | | | |
| 10 | - V | | | |

Plugging to an RREF solver (code attached):

Using Ohm's law

$$V_x = I_x R_x$$

	I	(mA)
Arc	1	136.363
B	2	-130.303
D	3	6.060
	4	251.732
E	5	-121.428
F	6	-60.606
G	7	7.142
H	8	196.969
	9	419.398
I	10	-92.857
J,L	11	128.571
K	12	221.428

Just means direction
assigned was wrong.
Take absolute value for
current answer.

V	(V)
A	1.363
B	1.303
C	1.363
D	0.303
E	1.214
F	3.030
G	0.214
H	1.969
I	2.785
J	1.285
K	2.214
L	1.285

- 3 - R with highest current
- 4 - I
- 5 - R with lowest current
- 6 - I
- 7 - R with highest voltage drop
- 8 - V
- 9 - R with lowest voltage drop
- 10 - V

K	221.428 mA
D	6.060 mA
F	3.030 V
G	0.214 V

- ① By KVL, the total voltage across series is sum of voltage drops, while current is constant.

$$V_{\text{tot}} = \sum V_i$$

$$I = I_1 = I_2 = \dots = I_n$$

Note that $V = IR$

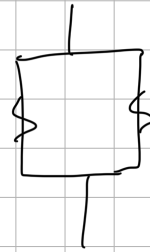
$$\text{so } I_{\text{tot}} R_{\text{tot}} = \sum I_i R_i$$

$$I R_{\text{tot}} = \sum I R_i = I \sum R_i$$

$$R_{\text{tot}} = \sum R_i$$

This generalizes to n resistors using induction.

- ② Note that parallel components have the same voltage



$$V = V_1 = V_2$$

$$I R_1 = I R_2$$

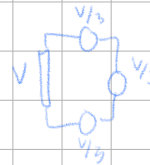
$$\text{so } I_1 = \frac{V}{R_1} \quad I_2 = \frac{V}{R_2}$$

while current splits

$$\text{By KCL: } I = I_1 + I_2$$

$$\frac{V}{R_{\text{eq}}} = \frac{V}{R_1} + \frac{V}{R_2}$$

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2} \Rightarrow R_{\text{eq}} = \left(\frac{1}{R_1} + \frac{1}{R_2} \right)^{-1}$$



Solution:

Start from $n/3$:



split this R into



split this $2R$ into



Our smallest resistor is $\frac{4R}{4}$.

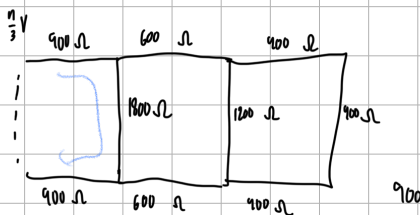
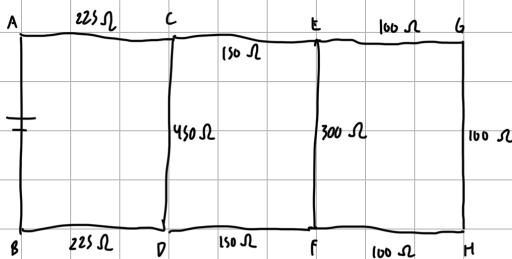
$$\text{so } \frac{4R}{4} = 100 \Rightarrow R = \frac{900}{4} = 225$$

$$\frac{2R}{3} = \frac{225(2)}{3} = 150$$

Hence:

$\frac{n}{3}$	AC, BD
$\frac{n}{4}$	CE, DF
$\frac{n}{27}$	EG, FH, FI

③



900Ω

$6 \rightarrow 2 \rightarrow$

2.25

1.5

1

$9 \rightarrow 4.5 \rightarrow 2.25$

4.5

3

1

$7.5 \rightarrow 3.7$

2.25

1.5

1

6

CS 20 Probset 2 Code Snippets

Diogn Lei R. Mortera

Contents

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1. Question 3

1.1. Code

```

import sympy as sp
from sympy import Matrix

raw = \
    """1 0 0 0 0 -1 0 -1 0 0 0 0 0
    0 1 0 1 1 0 0 0 0 0 0 0 0
    0 0 0 0 0 0 0 1 -1 0 0 1 0
    0 0 0 0 0 0 0 0 -1 1 -1 0
    0 0 0 0 -1 0 1 0 0 0 -1 0 0
    -1 -1 1 0 0 0 0 0 0 0 0 0 0
    0 0 -1 -1 0 1 -1 0 1 1 0 0 0
    -20 0 -50 0 0 -50 0 0 0 0 0 0 0
    0 10 50 0 0 0 0 0 0 0 0 0 -1
    0 0 0 0 -10 0 -30 0 0 0 0 0 1
    0 0 0 0 0 50 0 -10 0 0 0 0 -5
    0 0 0 0 0 0 30 0 0 30 20 0 0
    0 0 0 0 0 0 0 0 30 0 -10 -5
    """

def convert_string_to_matrix(raw):
    return [
        list(map(int, line.split()))
        for line in raw.strip().splitlines()
    ]

M = Matrix(convert_string_to_matrix(raw))

rref_matrix, pivot_cols = M.rref()
sp.pprint(rref_matrix)

A = M[:, :-1]
b = M[:, -1]

print("A shape:", A.shape)
print("rank(A):", A.rank())
print("nullity:", A.cols - A.rank())

print("rank([A|b]):", M.rank())

currents = []
for idx, val in enumerate(rref_matrix[:, -1], 1):
    currents.append(val)
    print(f"I_{str(idx).zfill(2)} = {'%.4f'%float(val*1000)} mA")

print(currents)

resistor_values = {
    'A': 10,
    'B': 10,
    'C': 10,
    'D': 50,
    'E': 10,
    'F': 50,
    'G': 30,
    'H': 10,
    'I': 30,
    'J': 10,
    'K': 10,
    'L': 10,
}

resistor_map = {
    'A': 1,
    'B': 2,
    'C': 1,
    'D': 3,
    'E': 5,
    'F': 6,
    'G': 7,

```



```

'H': 8,
'I': 10,
'J': 11,
'K': 12,
'L': 11,
}

for r, curr_idx in resistor_map.items():
    print(r, '%.4f'%float(abs(currents[curr_idx-1]*resistor_values[r])))

```

1.2. Output

```

[1 0 0 0 0 0 0 0 0 0 0 0 3/22 ]
|
|
| -43 |
| 0 1 0 0 0 0 0 0 0 0 0 0 0 — |
| 330 |
|
| 0 0 1 0 0 0 0 0 0 0 0 0 1/165 |
|
| 1163 |
| 0 0 0 1 0 0 0 0 0 0 0 0 0 — |
| 4620 |
|
| -17 |
| 0 0 0 0 1 0 0 0 0 0 0 0 0 — |
| 140 |
|
| 0 0 0 0 0 1 0 0 0 0 0 0 0 -2/33 |
|
| 0 0 0 0 0 0 1 0 0 0 0 0 0 1/140 |
|
| 13 |
| 0 0 0 0 0 0 0 1 0 0 0 0 0 — |
| 66 |
|
| 1933 |
| 0 0 0 0 0 0 0 0 1 0 0 0 0 — |
| 4620 |
|
| -13 |
| 0 0 0 0 0 0 0 0 0 1 0 0 0 — |
| 140 |
|
| 0 0 0 0 0 0 0 0 0 0 1 0 9/70 |
|
| 31 |
| 0 0 0 0 0 0 0 0 0 0 0 1 — |
| 140 |
|
|
| 0 0 0 0 0 0 0 0 0 0 0 0 0 0 ]
A shape: (13, 12)
rank(A): 12
nullity: 0
rank([A|b]): 12
I_01 = 136.3636 mA
I_02 = -130.3030 mA
I_03 = 6.0606 mA
I_04 = 251.7316 mA
I_05 = -121.4286 mA
I_06 = -60.6061 mA
I_07 = 7.1429 mA
I_08 = 196.9697 mA
I_09 = 418.3983 mA
I_10 = -92.8571 mA
I_11 = 128.5714 mA
I_12 = 221.4286 mA
I_13 = 0.0000 mA
[3/22, -43/330, 1/165, 1163/4620, -17/140, -2/33, 1/140, 13/66, 1933/4620, -13/140, 9/70, 31/140, 0]
A 1.3636
B 1.3030
C 1.3636
D 0.3030

```

E	1.2143
F	3.0303
G	0.2143
H	1.9697
I	2.7857
J	1.2857
K	2.2143
L	1.2857